Customer Data Analysis

PRDA-04 · Shopping behaviour in 10 Istanbul malls, 2021–2023

Power BI · Tableau · MySQL · Excel

About the dataset

- Source: MySQL database project_customer_segmentation, table customer
- One row per transaction: invoice, customer, gender, age
- Product category, quantity and amount paid (TL)
- Payment method, date and shopping mall
- Goal: who buys what, how they pay, and who brings revenue

99,457
transactions

10
shopping malls

Jan 2021 – Mar 2023
time period

Data checks and preparation

Clean data

No missing values, no duplicate rows or invoices

Price is already the total

price = quantity × unit price, so Revenue = price (68.55M TL)

One row per customer

Every customer_id appears once: count transactions, not repeat buyers

Partial last month

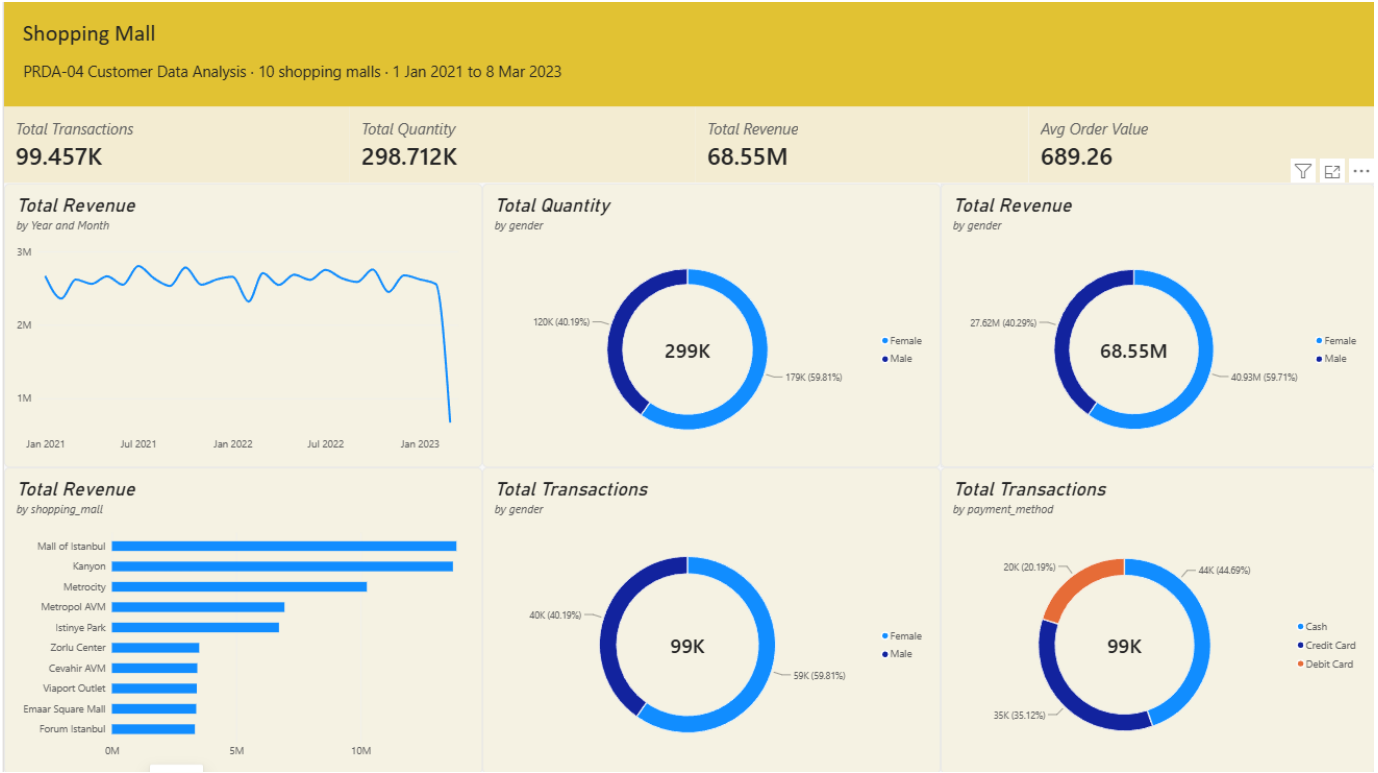
March 2023 has only 8 days of data

Key figures

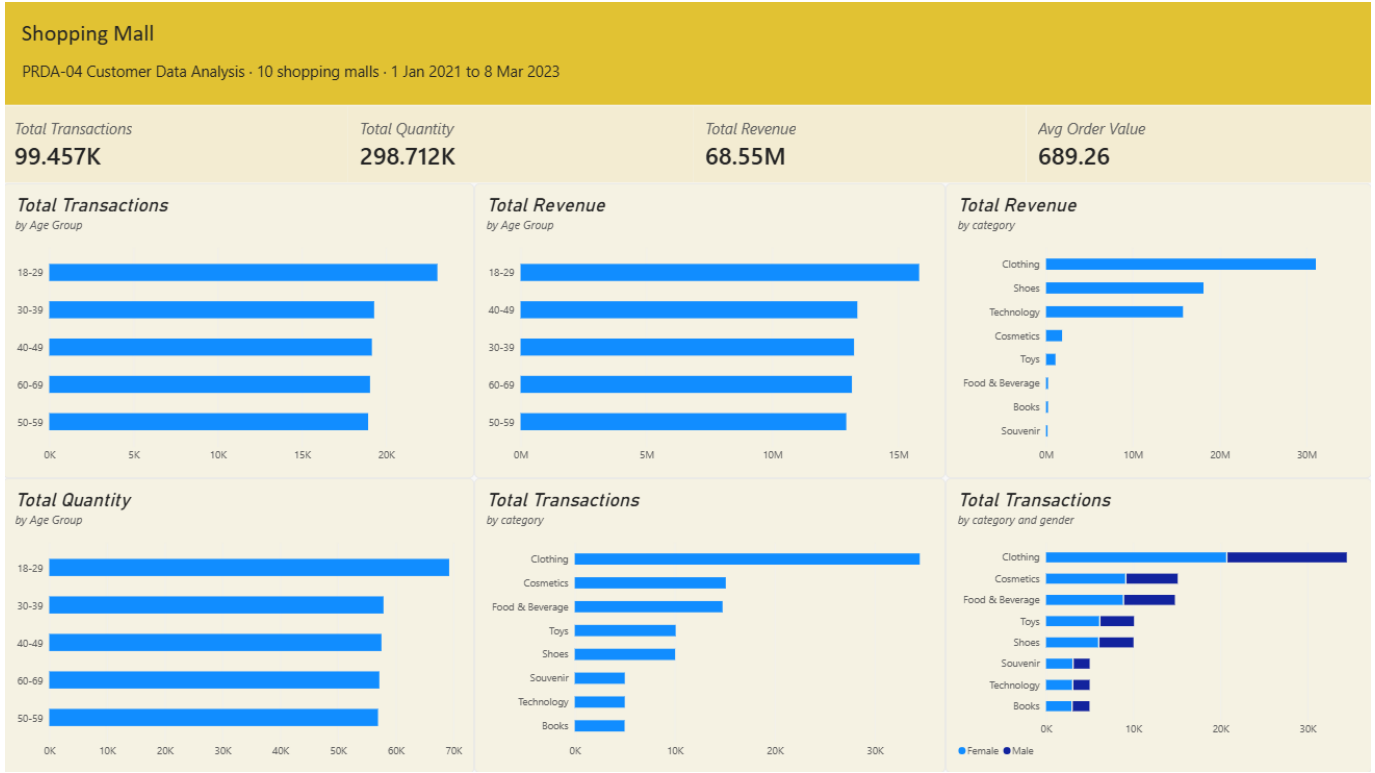
99,457	298,712	68.55M TL	689 TL
Transactions	Items sold	Revenue	Avg order value

3 items per purchase on average · average age **43.4** · about **2.6M TL** revenue per month

Power BI: overview, gender and payment



Power BI: age and category



Power BI: payment method relations

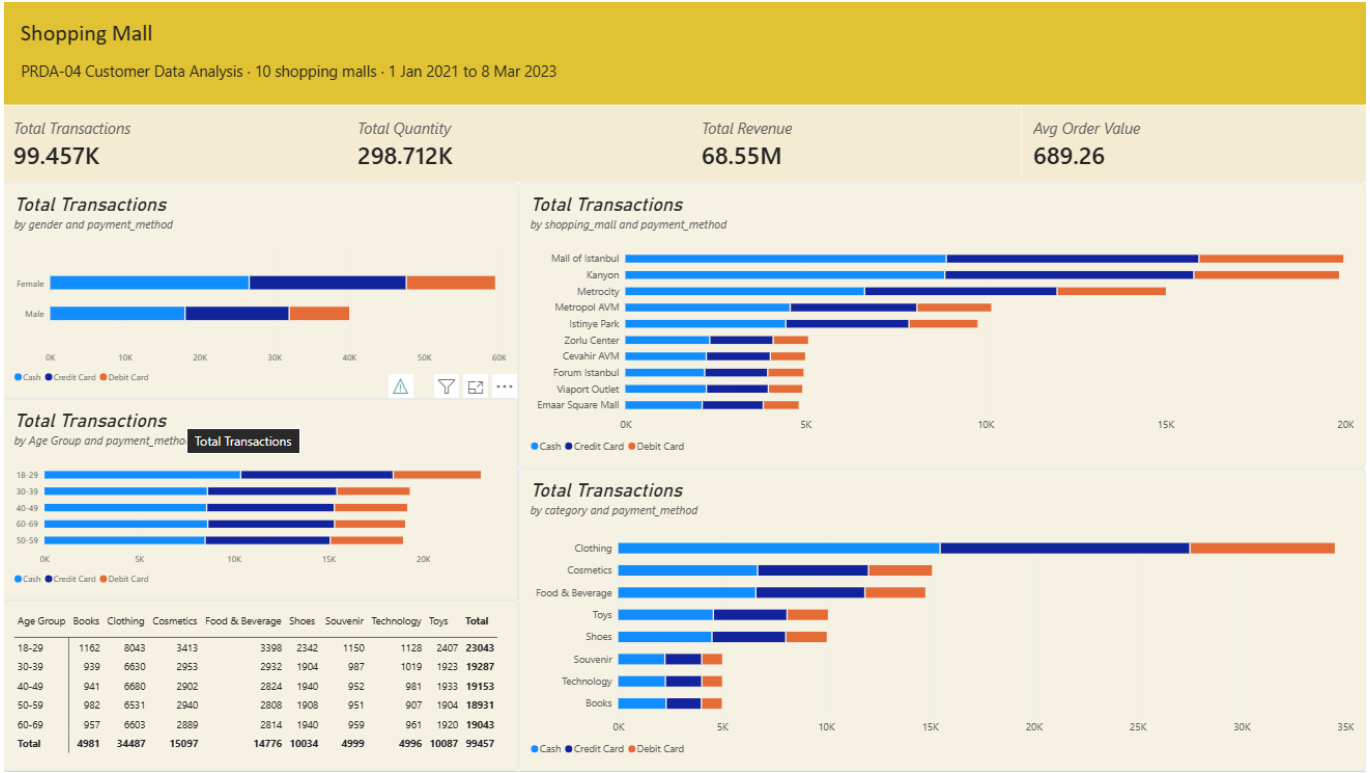


Tableau dashboard: page 1

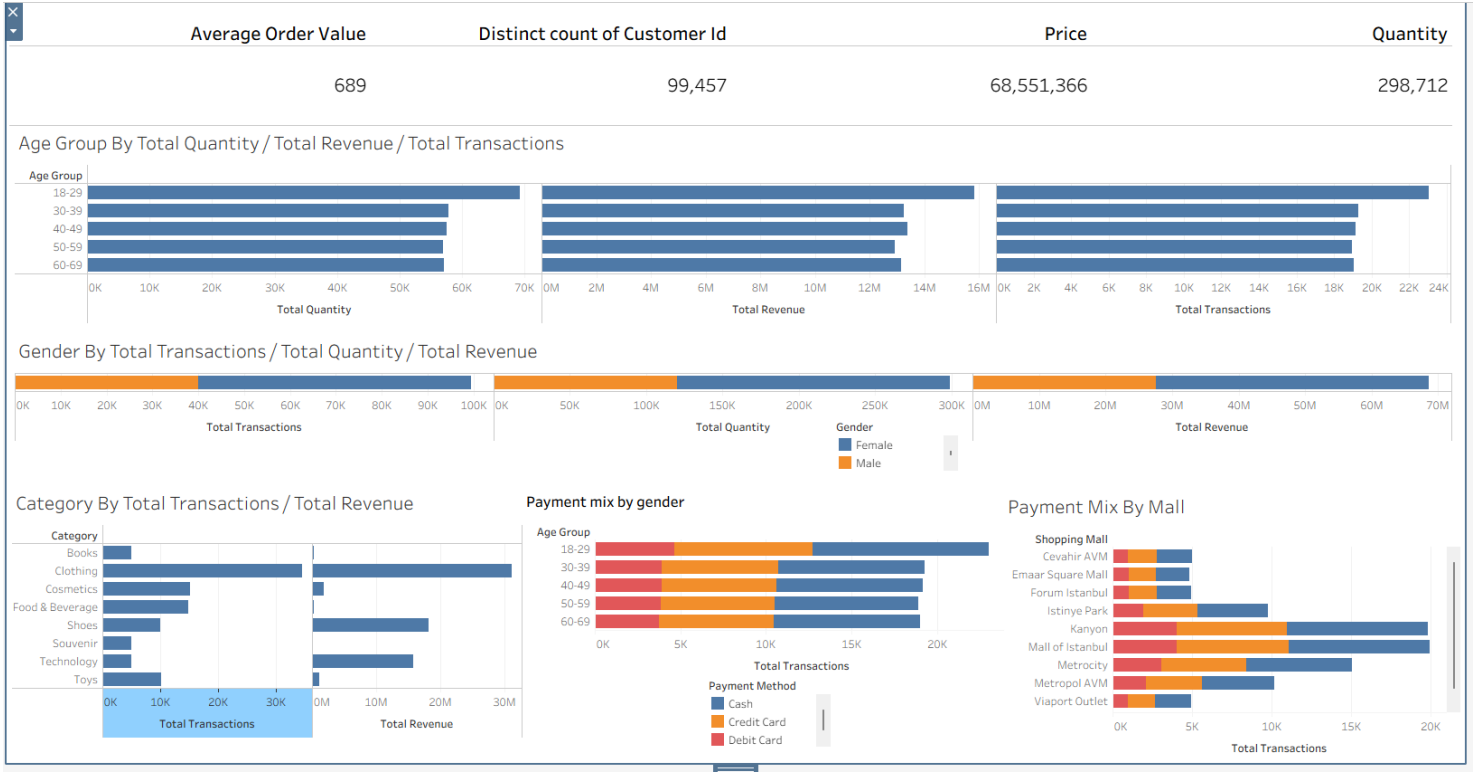
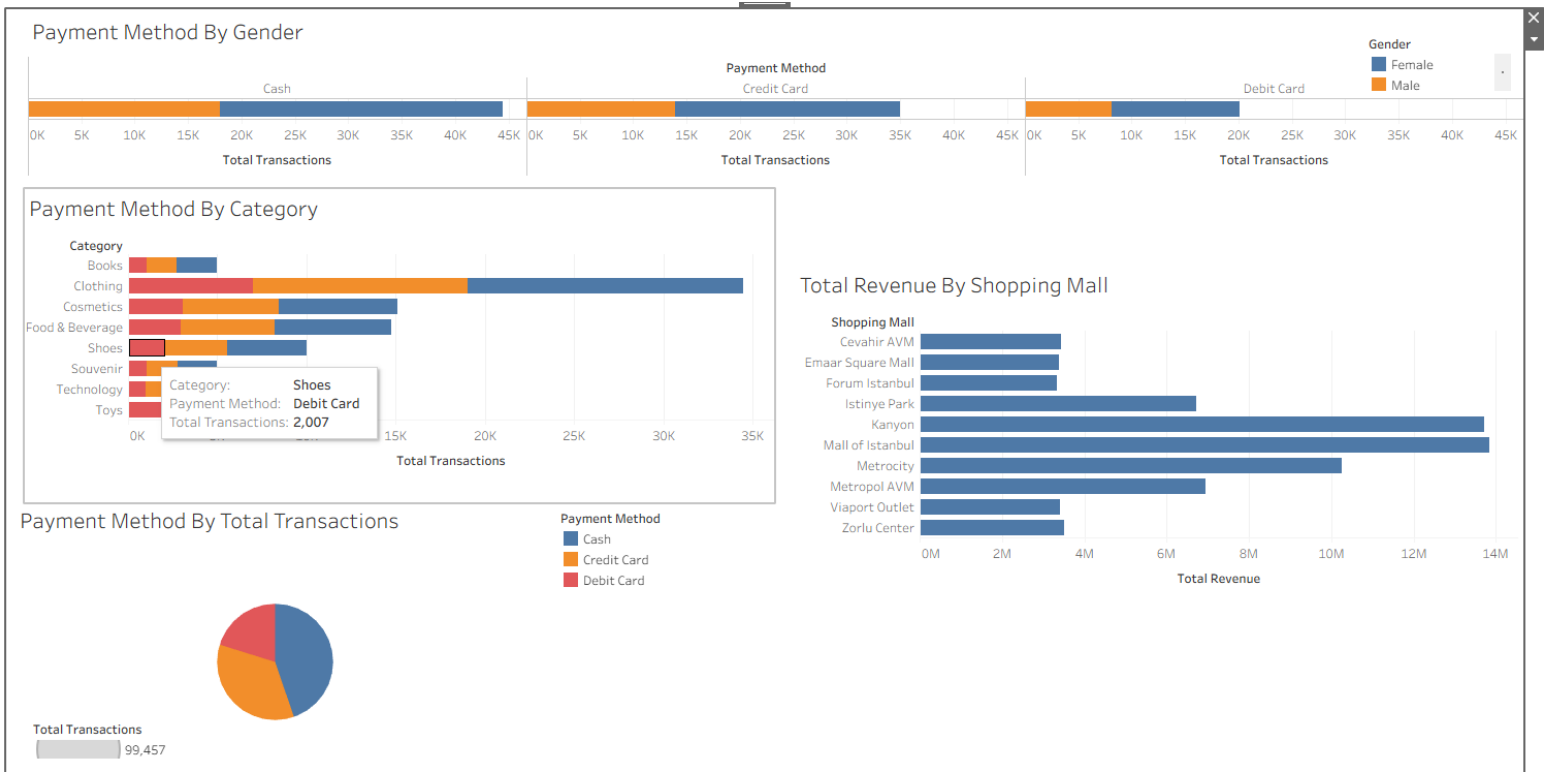
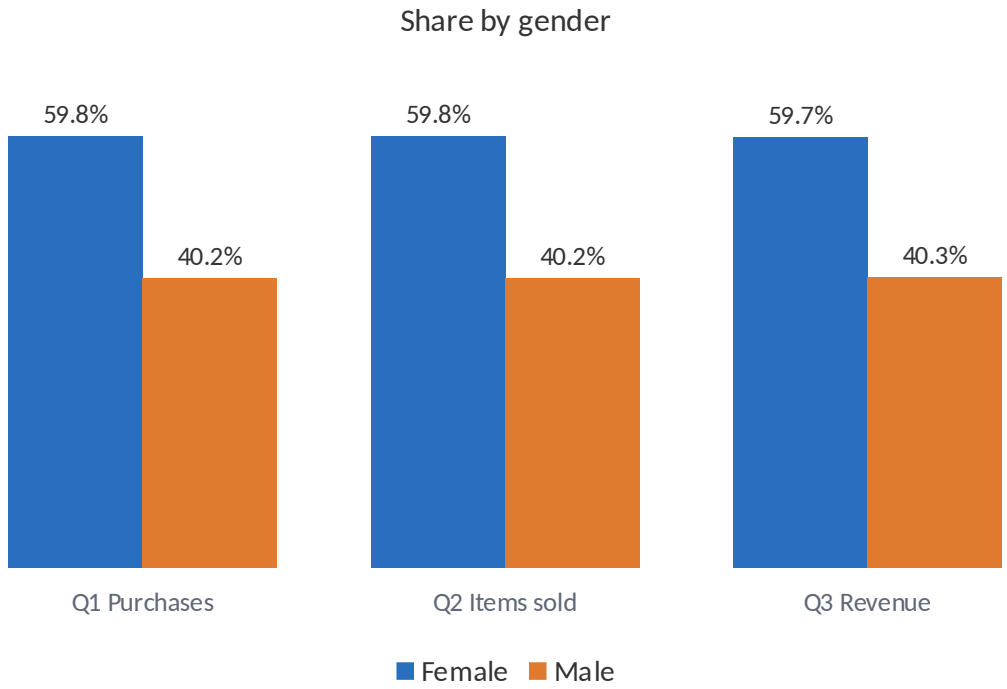


Tableau dashboard: page 2



Q1–Q3: Women lead on every measure

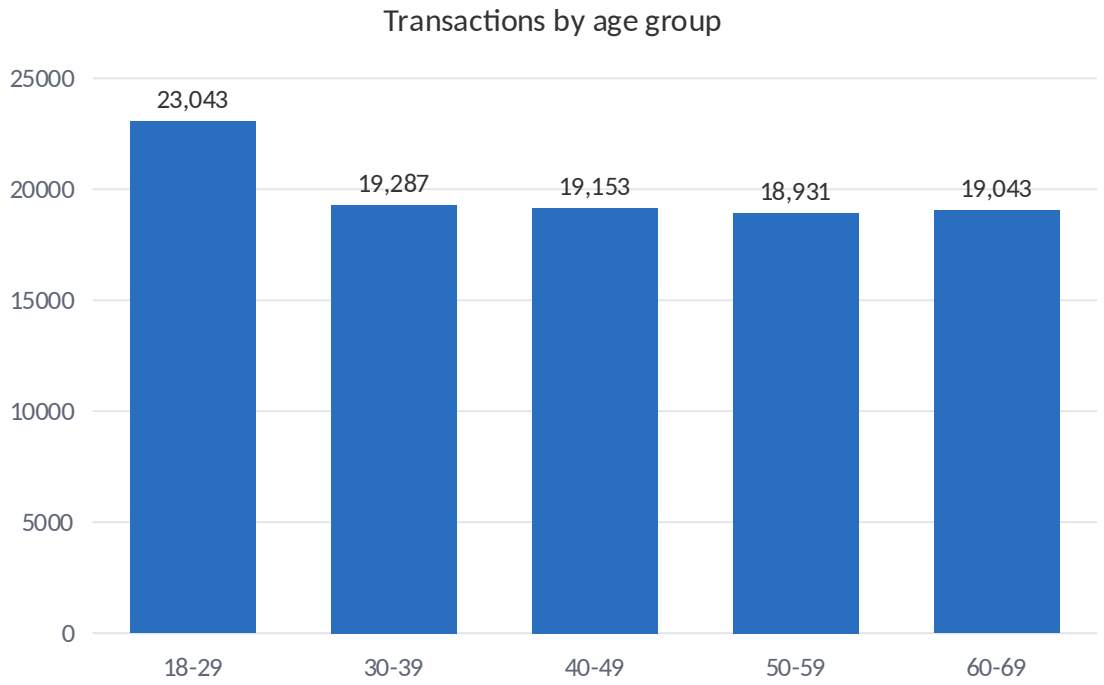


59.8%

of purchases made by women

Women bought 178,659 items and spent 40.93M TL. But the average order is the same for both (688 vs 691 TL): there are simply more female shoppers.

Q5–Q7: All ages shop about equally

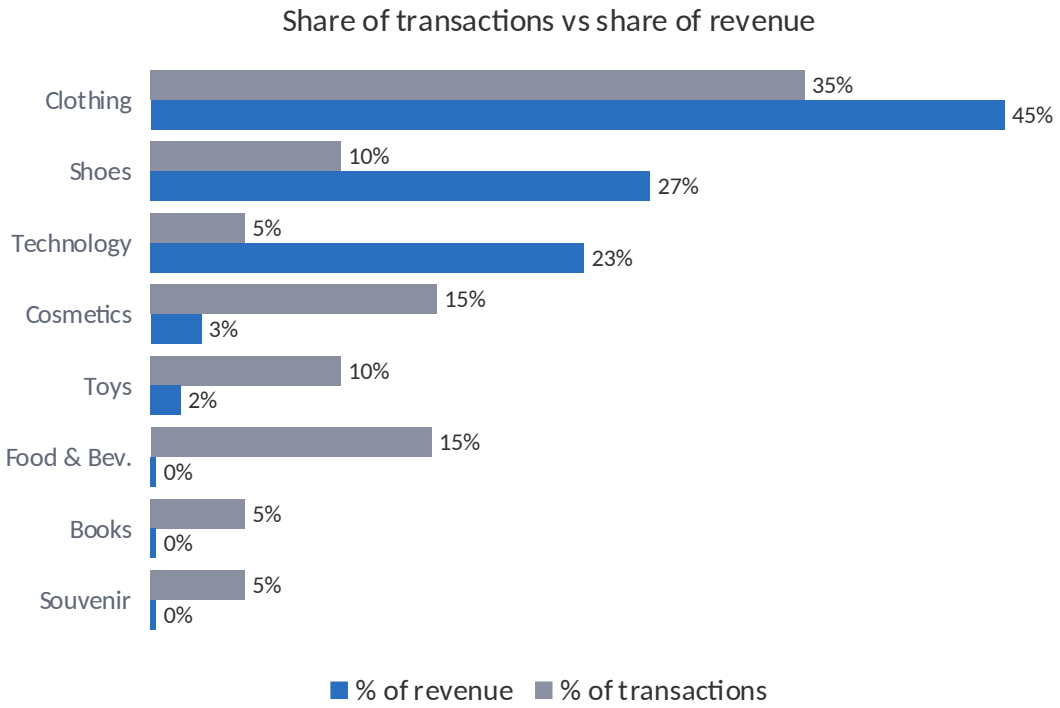


≈1,900

purchases at every single age

18-29 leads on purchases, items (69,245) and revenue (15.83M TL) only because it covers 12 ages instead of 10. Average order is 683–698 TL in every group.

Q4/Q8: Three categories bring 95% of revenue



94.8%

of revenue from Clothing, Shoes, Technology

Food & Beverage is 15% of purchases but only 0.3% of revenue: one item costs 5.23 TL, compared with 1,050 TL for Technology.

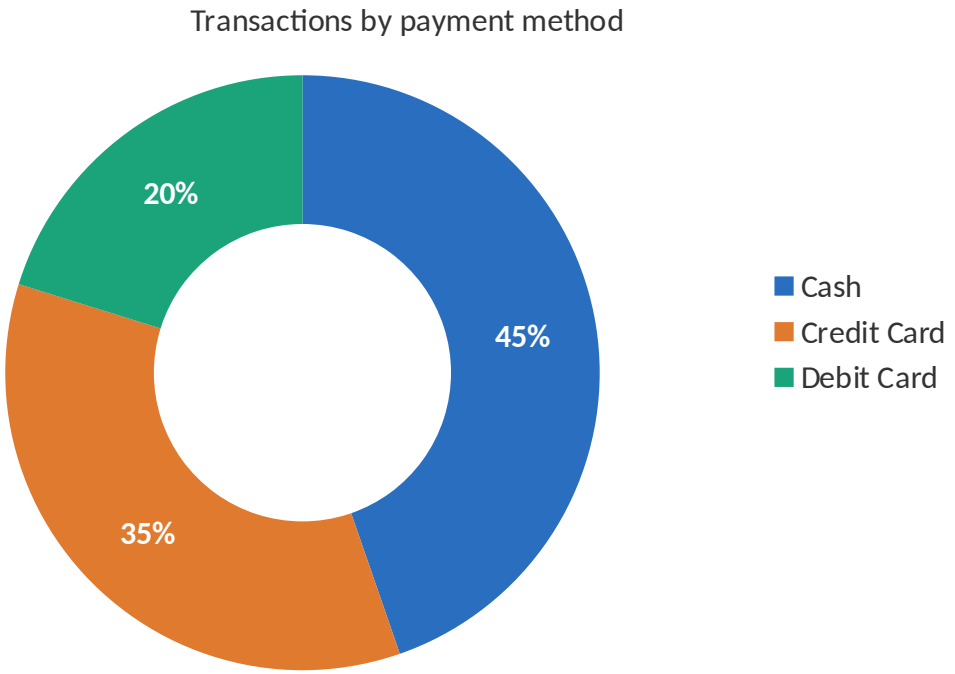
Q4/Q8: Every group buys the same mix

Age group	Clothing	Cosmetics	Food & Bev.	Shoes	Toys	Books	Souvenir	Tech
18-29	34.9%	14.8%	14.7%	10.2%	10.4%	5.0%	5.0%	4.9%
30-39	34.4%	15.3%	15.2%	9.9%	10.0%	4.9%	5.1%	5.3%
40-49	34.9%	15.2%	14.7%	10.1%	10.1%	4.9%	5.0%	5.1%
50-59	34.5%	15.5%	14.8%	10.1%	10.1%	5.2%	5.0%	4.8%
60-69	34.7%	15.2%	14.8%	10.2%	10.1%	5.0%	5.0%	5.0%

% of each age group's purchases. Each category's share changes by less than 1 point between age groups.

Gender: every category is 58–60% female. **Mall:** Clothing is 34–35% of purchases in every mall.

Q10: Cash is the most used payment method

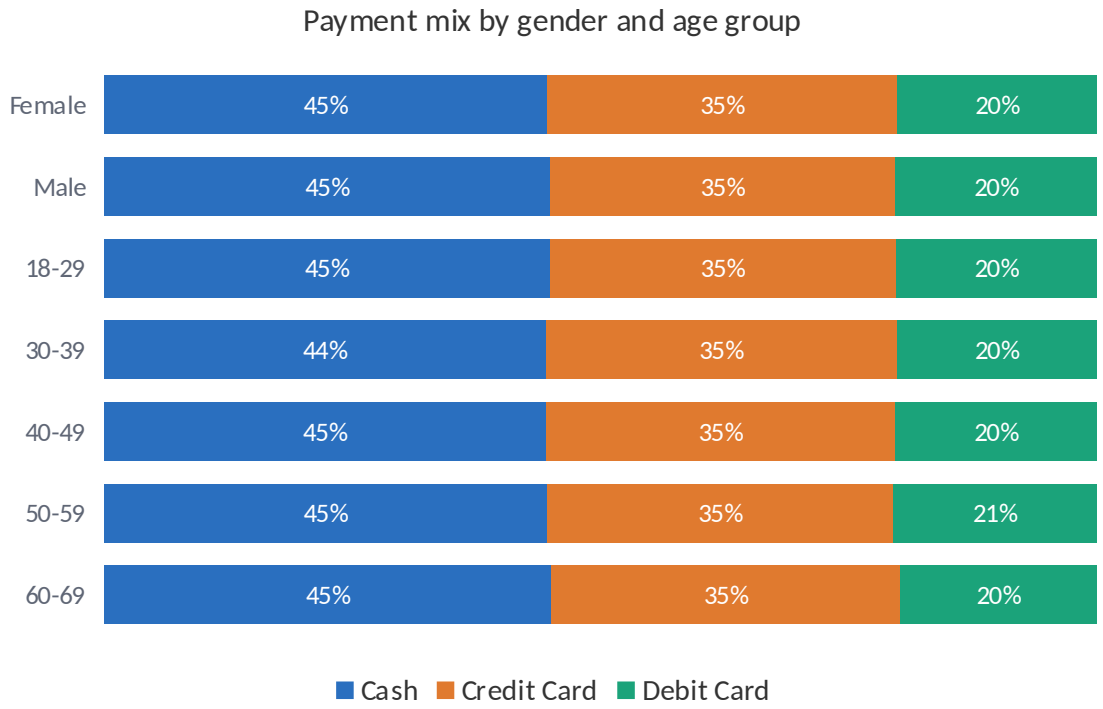


55.3%

of purchases paid by card

Cash 44,447 (44.7%), Credit Card 34,931 (35.1%), Debit Card 20,079 (20.2%).

Q9: Payment mix is the same for everyone

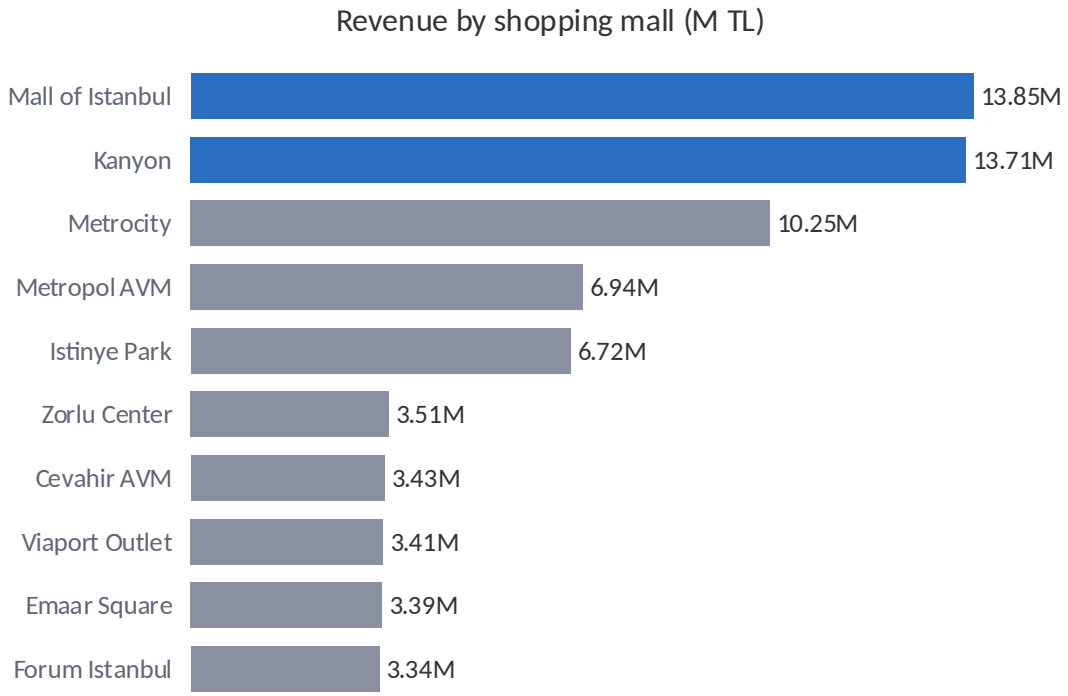


45 / 35 / 20

Cash / Credit / Debit in every group

The same split holds for every category and mall too (within about 1.5 points), and the average order is 687–691 TL for each method.

Two malls bring 40% of revenue



27.6M TL

from Mall of Istanbul and Kanyon

Five malls bring only about 3.4M TL each. Monthly revenue is steady at about 2.6M TL, with a small dip every February.

Key insights

Women are 60% of sales

Same order value as men (about 690 TL)

All ages shop equally

About 1,900 purchases at every age

Clothing is the leader

45% of revenue from one category

Revenue is concentrated

3 categories = 95%, 2 malls = 40%

Cash is most used

45% cash, 55% card

No hidden patterns

Category and payment mix same for all groups

Suggestions for the company

1. **Focus on top categories**
   Clothing, Shoes and Technology: stock, space, promotions

2. **Cross-sell low-price items**
   Food and cosmetics as bundles that lead to clothing

3. **Grow male shoppers**
   Men spend as much per order: target campaigns

4. **Market to all ages**
   Use age to choose channels, not offers

5. **Start a loyalty programme**
   Track repeat visits and give personal offers

6. **Encourage card payments**
   Faster checkout and better customer data

7. **Learn from top malls**
   Apply Mall of Istanbul and Kanyon ideas elsewhere

8. **Plan for February**
   Seasonal campaign to fill the yearly dip

Conclusion

99,457 sales worth 68.55M TL with steady monthly revenue. Customer groups behave alike; revenue depends on three categories and two malls. Protect those, grow male shoppers and start a loyalty programme.